

Kartavya Suryawanshi

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EDUCATION

Indian Institute of Technology, Mandi

Aug 2023 – Present

B.Tech in Data Science

CGPA: 9.27/10.0

- **Relevant Coursework:** Design of Algorithms, Data Handling and Visualization, Machine Learning, Deep Learning, Computer Organization, Computer Systems for Data Science, Probability and Statistics, Matrix Computation for DS

TECHNICAL SKILLS

- **Core CS Fundamentals:** Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP), System Design, Distributed Systems, Problem Solving
- **Languages:** C/C++, Python, Java, JavaScript, TypeScript, SQL, Bash
- **Frameworks, APIs & Cloud:** React, Next.js, FastAPI, Django, REST APIs, PyTorch, TensorFlow, Docker, Kubernetes, AWS, PostgreSQL, Redis
- **Tools, Practices & Applied AI:** Git, Linux, CI/CD, Agile/Scrum, GitHub Copilot, LaTeX, Prompt Engineering, RAG

WORK EXPERIENCE

Nomura Real Estate Holdings, Inc. [Certificate]

June 2026 – July 2026

Data Analytics Intern

Tokyo, Japan

- Engineered scalable analytics dashboards over 10,000+ Microsoft Viva Insights and GitHub Copilot logs for AI adoption.
- Conducted statistical analysis and hypothesis testing on work patterns, delivering 3 company-wide AI adoption strategies.

IIT Mandi Finance Department [Letter of Recommendation]

Aug 2025 – Dec 2025

Full-Stack Developer

Mandi, India

- Owned SDLC for a secure, role-based finance platform using Next.js, TypeScript, Supabase and NextAuth.js with RBAC.
- Shipped reliable approval workflows for billing, auditing and procurement, serving 50+ staff and ₹10+ lakh in transactions.

PROJECTS

Anatomy-Aware DoseFlow | PyTorch, MedSAM, ViT, Mamba

Feb 2026

- Created a three-stage deep learning pipeline for reconstructing CT images from arbitrary dose levels (5%–100%).
- Devised a dose-conditioned flow trajectory model integrating MedSAM, ViT encoders and four loss functions.
- Achieved 48.15 dB PSNR and 0.9991 SSIM with zero-shot generalization across three unseen anatomical regions.

FLOW — Fraud and Loan Optimization Workbench | Pathway, NATS JetStream, Redis, FastAPI, React

Nov 2025

- Architected event-driven, distributed ML pipelines for real-time fraud detection using Pathway, Redis and NATS JetStream.
- Deployed adaptive online-learning models (Hoeffding Trees, Online GMM) with human-in-the-loop feedback.
- Attained sub-5 ms latency and 88% F1-score, instrumenting Prometheus and Grafana for observability.

LunaDEM — Lightweight Lunar Terrain Analysis | Python, NumPy, SciPy, Matplotlib

Oct 2025

- Published a lightweight Python library for 3D lunar terrain analysis using NumPy and SciPy with zero-configuration setup.
- Programmed 4 DEM reconstruction algorithms to generate 3D lunar topography from 2D orbital imagery.
- Formulated rover-aware landing site evaluation using terrain slope, roughness, and curvature for 9+ real-world rovers.

LokMitra AI | Django, Next.js, Twilio, Qwen2-Audio, PostgreSQL

Sept 2025

- Engineered a multilingual voice AI platform (Twilio, Qwen2-Audio) using prompt engineering for audio-text reasoning.
- Orchestrated a ReAct-based agentic system with RAG, database querying, tool calling and human handoff.
- Configured asynchronous, cloud-native webhooks for audio streaming, transcript processing and call orchestration.

AWARDS & ACHIEVEMENTS

- **1st Place** (Deepfake Track), **Hack 60 by HCLTech** — Designed VoiceGuard, an anti-spoofing AI model.
- **1st Place**, **NASA Space Apps** (Chandigarh) — Authored Astrogenesis, an LLM research copilot.
- **1st Place Overall** (1,600+ teams), **iHub Multimodal Hackathon '25** — Led 5-member *Smarts Scribe* team.
- **Runner-Up** (Autonomous Agents), **FrostHack IIT Mandi** — Spearheaded AutoReach, an automated workflow agent.

LEADERSHIP & VOLUNTEERING

- **Core Member, Programming Club** — Conducted 15+ algorithms workshops impacting 300+ students.
- **Technical Lead, STAC** — Owned technical initiatives, mentored student research team, and organized events.